



SkyGeni Practicum

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1. Project Scope and Objectives

SkyGeni partnered with our practicum team to build a sequential purchase recommender that predicts each B2B account's next product purchase based on historical order data. The core business question is direct: given everything an account has already bought, what should be offered to them next? A reliable answer to this question drives cross-sell and upsell campaigns, prioritizes accounts for sales outreach, and sharpens forecasting at the product-family level.

Two design choices shape the formulation. First, we model purchases as an ordered sequence rather than an unordered basket — order matters because the product typically bought after a customer's first deal differs from the one purchased after their fifth. Second, the pipeline is granularity-tunable, supporting Product_Family, Product_Group, or Product_Name. The deliverable in scope was generated at Product_Name granularity (the most useful for sales enablement, but also the sparsest).

Project objectives:

- Build a reproducible pipeline that learns $P(\text{next product} \mid \text{prior sequence})$ from purchase history.
- Combine multiple complementary signals (Markov chains, association rules, stage-aware popularity) into a single hybrid scorer.
- Evaluate end-to-end with a production-realistic protocol (leave-last-out).
- Deliver actionable top-K recommendations with full per-signal score decompositions for stakeholder explainability.

2. Work Plan and Timeline

The project followed a five-phase plan. Each phase produced a checkpoint deliverable reviewed with the SkyGeni sponsor before advancing.

Phase	Focus	Activities & Checkpoint Deliverable
1	Discovery & data audit	Reviewed Purchase_Data.csv (33,589 events; 3,981 accounts; 634 products; Mar 2020–Mar 2026). Confirmed schema, granularity options, and evaluation expectations with sponsor.
2	Pipeline & MarkovChain class	Built data loader, deduplication, sequence construction, and leave-last-out splitter. Implemented a reusable Nth-order MarkovChain class supporting orders 1–4.
3	Signal expansion	Added association rules (confidence and lift) and stage-aware popularity. Designed the hybrid scorer with a cascading fallback so unseen higher-order states do not lose their weight.
4	Evaluation & ablation	Ran leave-last-out across 3,510 eligible accounts. Computed Hit@K, MRR@K, NDCG@K. Ran ablation isolating each of the seven signals to quantify their individual contributions.



Phase	Focus	Activities & Checkpoint Deliverable
5	Reporting & handoff	Produced two Jupyter notebooks, an Excel workbook with six sheets (Cover, Evaluation, Ablation, Coverage, Recommendations, Methodology), and a technical report with code walkthrough, metric glossary, and operational guide.

3. Key Project Findings and Deliverables

Headline result: for 68% of accounts the recommender's #1 pick is exactly the next product they buy, and for over 98% the correct product appears somewhere in the top 10. *The hybrid model substantially outperforms any single signal in isolation — the strongest solo signal (first-order Markov) reaches only Hit@10 = 78%.*

K	Hit@K	MRR@K	NDCG@K	N evaluated
1	68.29%	0.6829	0.6829	3,510
3	85.70%	0.7585	0.7837	3,510
5	93.70%	0.7769	0.8168	3,510
10	98.26%	0.7832	0.8318	3,510

Why the hybrid works: it combines seven signals — four Markov chains (orders 1–4), two association-rule measures (confidence and lift), and stage-aware popularity. A cascading fallback redirects weight from unseen higher-order states down to lower orders so no signal weight is wasted. This design lifts Hit@10 by roughly 20 percentage points over the best solo signal.

Deliverables provided to SkyGeni:

- **Two Jupyter notebooks** (SkyGeni-v2-multiplesteps.ipynb, SkyGeni-v3-multiplesteps.ipynb) implementing the data pipeline and the reusable MarkovChain class.
- **An Excel workbook** (sequential_recommender_results_V2_equal_weights.xlsx) with six sheets: Cover, Evaluation, Ablation, Coverage, Recommendations (top-10 per account, 35,100 rows with full per-signal score decomposition), and Methodology.
- **A technical report** with code walkthrough, metric glossary (Hit@K, MRR@K, NDCG@K, support/confidence/lift), design-decision notes, an operational guide for re-running and tuning the pipeline, and a business-readers' glossary.



4. Benefits to SkyGeni

- **Production-ready next-purchase predictions.** With $\text{Hit}@5 \approx 94\%$, sales teams receive a five-product shortlist that almost always contains the right next move — directly usable for cross-sell and upsell prioritization.
- **Explainability built in.** Every row in the Recommendations sheet exposes the seven per-signal scores, so account managers can answer "why this product?" using the two or three signals that actually drove the ranking.
- **Deterministic and cheap to retrain.** The model is a counting model — identical input yields identical output, with no random initialization. A monthly retrain is a safe default; weekly is feasible because retraining is a single counting pass, not gradient descent.
- **Portable across catalogs.** Product codes are obfuscated and treated as opaque identifiers, so the pipeline transfers to any future SkyGeni client catalog with no code changes.
- **Tunable to business rhythm.** Granularity, sequence-length filters, support thresholds, top-K size, and signal weights are all exposed as configuration; the operational guide documents when to turn each knob.

5. Proposed Next Steps

Three concrete extensions are recommended, in priority order. Each is annotated with order-of-magnitude impact and effort estimates rather than commitments.

- **Weight tuning (low effort, ~1–2 days).** A 3-fold validation grid over the WEIGHTS dictionary, using a held-out validation slice kept separate from the leave-last-out evaluation set. Expected gain: +1 to +3 points on $\text{Hit}@5$. Risk-free first step because the evaluation harness already exists.
- **Time decay in the Markov fit (medium effort, ~4–5 days).** Apply an exponential decay (e.g., one-year half-life) to transition counts so 2020 purchases weigh less than 2026 ones. Expected gain: +2 to +5 points if the catalog has meaningful churn. Implementation is a single multiplicative weight inside the counting loop; tuning the half-life is where the time goes.
- **Cohort ensembling (high effort, 2+ weeks).** Cluster accounts by industry $\times$ size, fit one MarkovChain per cohort, and blend predictions by cohort similarity at inference. Expected gain: +3 to +8 points on cold-start accounts, with smaller gains on mature accounts. Highest ceiling, most moving parts.

Beyond these three, the natural longer-horizon research direction is a neural sequential model (GRU4Rec, SASRec) that can blend account- and product-level features with the sequence signal already captured here. The current Markov-based hybrid is a strong baseline against which any such extension should be measured.

